Numerical Methods and Optimization

20/4/2026 – 3^{rd} Year

ENSIA — Spring 2026

Tutorial 6

- Polynomial Interpolation -

Exercise 1: Identification

We consider $x, y \in \mathbb{R}^4$ given by: $x = [-2, 0, 1, 2]$ and $y = [4, 0, 0, 4]$. Among the following polynomials, which one is the interpolation polynomial P at the points x, y (justify your answer)?

1. $P_1(X) = X^4 - \frac{2}{3}X^3 - 3X^2 + \frac{8}{3}X$
2. $P_2(X) = \frac{3}{4}X^2 - \frac{4}{3}$
3. $P_3(X) = \frac{1}{3}X^3 + X^2 - \frac{4}{3}X$

Exercise 2: Existence and Uniqueness

1. Show that there are infinitely many polynomials of degree 2 whose graph passes through the points $(0, 0)$ and $(1, 0)$.
2. Show that there does not exist a polynomial of degree 2 passing through the points $(0, 1)$, $(1, 4)$, $(2, 15)$, and $(3, 40)$.

Exercise 3: Construction... Smart or Rough approach

Compute the Lagrange interpolation polynomials for the following points:

a. $x = [-1, 2, 3]$ and $y = [4, 4, 8]$

b. $x = [-2, -1, 0, 1]$ and $y = [0, -2, -4, 0]$

c. $x = [-1, 0, 1, 2]$ and $y = [6, 2, 0, 0]$

Exercise 4: Using the Characterization

Let P be a polynomial. Show that its interpolation polynomial at the nodes $x_i \in \mathbb{R}$, $0 \leq i \leq n$, is the remainder of the Euclidean division of P by the polynomial $\pi_n(x) = (x - x_0)(x - x_1)\dots(x - x_n)$.

Exercise 5: Newton polynomial

(1) Recall the expression of the Newton basis of $\mathbb{R}_5[X]$ associated with the nodes $1, 2, 3, 4, 5, 6$.

(2) Show that it is indeed a basis.

(3) Give, in this basis, the expression of the polynomial $P \in \mathbb{R}_5[X]$ such that:

$$P(1) = P(6) = 1 \quad \text{and} \quad P(2) = P(3) = P(4) = P(5) = 0.$$

Exercise 6: Lagrange's plynomial

(1) Compute the Lagrange interpolation polynomial of the function $f(x) = x(x^2 - 1)$ with respect to the points $x_0 = -1, x_1 = 1, x_2 = 2$. Same question, adding the point $x_3 = -2$.

Exercise 7: Interpolation error

Let $f(x) = \cos(x)$, for $x \in [0, 1]$. Estimate the minimum number of interpolation nodes such that the error between f and its Lagrange interpolation polynomial is less than 0.01.

Exercise 8: Smart Ad Recommendation on Social Media

A social media platform uses an AI system to predict how likely a user is to click on an advertisement depending on the time spent scrolling before seeing the ad.
The following historical data was collected:

Time spent scrolling x (seconds)	Click probability y
1	0.12
2	0.20
4	0.35
5	0.42

The AI needs to estimate the click probability for users who spent times not present in the dataset.

(1) Construct the interpolation polynomial using the Lagrange method.

(2) Construct the interpolation polynomial using the Newton divided differences method.

(3) Estimate the probability of clicking when:

- $x = 3$ seconds
- $x = 4.5$ seconds

(4) Compare the interpolated values with linear interpolation between nearest points.

(5) Explain why interpolation can help when AI systems have limited training data.

Interpretation:

- If the predicted probability is high, the platform shows premium ads.
- If the probability is low, the platform shows cheaper ads or skips the ad.

Exercise 9: Rolle's theorem

(1) Let the function $f(x) = sin(x)$, and let $P_2(x)$ the interpolation polynomial of f at the points: $x_0 = 0, x_1 = \frac{\pi}{4}, x_2 = \frac{\pi}{2}$. Bound the error $|f(x) - P_2(x)|$ for $x = \frac{\pi}{5}$.